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22PHYS12

B.E. First Semester Examination – January 2025

Physics for CSE Stream

(Common to CSE, CSD, ISE and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Physical constants:
Velocity of light, $C = 3\times10^{8} ms^{-1}$: Planck's constant, Boltzmann constant, $K = 1.38\times 10^{-23} J/K$:
Avagadaro number $N_A = 6.023 \times 10^{26} qm/kmole$;
Permittivity of Vacuum, $\varepsilon_o = 8.85\times10^{-12} Fm^{-1}$: Mass of Neutron, $= 1.67 \times10^{-27} kg$,
Acceleration due to gravity $g = 9.8\ ms^{-2}$.

Module – I

1. a) Illustrate the construction and working of (GaAs) semiconductor diode laser with suitable diagram. **(09 Marks)**
   b) Derive the expression for numerical aperture and arrive the condition for propagation of light in an optical fibre. **(08 Marks)**
   c) Find the ratio of population of two energy level in a medium in thermal equilibrium, if the wavelength of light emitted at 330K is 632.8 nm. 1.05×10^{-30} **(03 Marks)**

2. a) With neat diagram explain the different types of optical fibers. **(09 Marks)**
   b) Explain bar code scanner with neat diagram. **(07 Marks)**
   c) The attenuation of light in an optical fiber is estimated at 2.2 dB/km. What fractional initial intensity remains after 2 km and 6 km? 0.363 & 0.0479 **(04 Marks)**

Module – II

3. a) Develop on dimensional time-independent schrodinger wave equation. **(09 Marks)**
   b) Explain Heisenberg's uncertainity principle. Give its physical significance. **(07 Marks)**
   c) Calculate the zero point energy for an electron in a box of width 10 Å. 0.37 eV **(04 Marks)**

4. a) Illustrate the non-existence of electrons in the nucleus on the basis of beta-decay observation using Heisenberg uncertainty principle. **(08 Marks)**
   b) Derive an expression for de-Broglie wave length using the concept of group velocity. **(08 Marks)**
   c) A particle of mass 0.65 MeV/c^2 has free energy 120 eV. Find its de Broglie wavelength. C is the velocity of light. $\lambda = 9.95 \times 10^{-11} m$ **(04 Marks)**

Module – III

5. a) Explain in detail polarization mechanisms in dielectrics. **(10 Marks)**
   b) Explain the frequency dependence of polarization with neat sketch. **(06 Marks)**
   c) A parallel plate capacitor of area 650 mm^2 and a plate separation of 4 mm has a charge of $2 \times 10^{-10} C$ on it. What is the resultant voltage across the capacitor when a materials of dielectric constant 3.5 is introduced between the plates. $V = 39.73 V$ **(04 Marks)**

6. a) Derive an expression for internal field in case at one dimensional array of atoms in dielectric solids. **(09 Marks)**
   b) Derive Clausius-Mossotti equation. **(08 Marks)**
   c) The electronic polarization of Krypton gas is $3.54 \times 10^{-40} Fm^2$.
   If gas contain $2.7 \times 10^{25}/m^3$ at NTP, calculate its dielectric constant. $\varepsilon_r = 1.00108$ **(03 Marks)**

Module – IV

7. a) Obtain an expression for density of states for energy level in a band. **(10 Marks)**
   b) Describe the experimental procedure to find the Planck's constant using an LED. **(10 Marks)**

8. a) Explain type I and type II superconductors. (08 Ma
   b) The Fermi level in silver is 5.5 eV. What are the energies for which the probabilities of occupancy at 300 K is 0.99. (Given $1J = 6.24 \times 10^{18}eV$). (05 Marks)
   c) Calculate the wavelength of laser beam, if the distance between the grating element and screen is 150 cm and the first order and second order diffraction maxima is 10 cm and 20 cm (Given N = 2500 lines per inch (LPI)) $\lambda_1 = 6760$, $\lambda_2 = 6739\text{Å}$ (07 Marks)

Module – V

9. a) Discuss the sailent features of normal distribution using Bell curves. (06 Marks)
   b) Explain the strides and steps in ability to walk. (04 Marks)
   c) Determine the dielectric constant of capacitors using RC circuit. If the length and breadth of capacitor is 47 mm and 5 mm and separation between them is 0.075mm given R = 100 kΩ. (10 Marks)

Time in seconds	Charging	Discharging
0	0	10.3
10	6.1	4
20	8.6	1.6
30	9.6	0.6
40	10	0.2
50	10.2	0.1
60	10.3	0.0

$K = 3.89$

10. a) Explain description and inferential statistics. (05 Marks)
    b) Explain character animation of jumping with its parts of jump. (05 Marks)
    c) Calculate the Fermi energy of copper wire for a given data below with constant 5×10^{-13}. (10 Marks)

Temperature in K	Voltage mV	Current mA
303	20.9	47
308	21.2	47
313	21.5	47
318	21.9	46.9
323	22.2	46.9
328	22.5	46.9

$E_F = 3.125\,eV$